

Leveraging Edge Intelligence for Solar Energy Management in Smart Grids

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ABSTRACT The rapid advancement of renewable energy, particularly solar power, has been significantly supported by the Internet of Things (IoT) infrastructure, enabling optimized energy management in smart grid systems. However, accurate long-term solar energy forecasting remains a critical challenge, especially exploiting specific characteristics of data locality at the edge. This paper introduces an edge intelligence-driven hybrid deep learning model that integrates Temporal Convolutional Networks (TCN) and Gated Recurrent Units (GRU) for precise solar energy prediction. The proposed model leveraged TCN's capability to capture long-range temporal dependencies and GRU's efficiency in sequential data processing. To optimize performance for edge computing environments, we employed Neural Architecture Search, model pruning, and quantization techniques, reducing both model size and complexity. The model has been validated using two benchmark datasets (Alibaba Competition and GEFCom), achieving an impressive mean squared error of 0.0051 on long-term forecasts. The results demonstrate that our approach ensures high forecasting accuracy while maintaining computational efficiency, making it well-suited for real-time, IoT-enabled solar energy applications in smart grids.

INDEX TERMS Internet of Things, solar energy forecasting, hybrid deep learning, edge intelligence

I. INTRODUCTION

The increasing global demand for sustainable energy solutions has accelerated the integration of renewable energy sources into modern power systems. Among these, solar energy has emerged as a pivotal component due to its abundance and environmental benefits [1]. However, its integration into modern power systems poses unique challenges due to its intermittent and variable nature. Fluctuations in solar power generation arise from local factors such as weather conditions, seasonal variations, and geographic location, making it difficult for management [2], [3]. Efficient solar energy forecasting is essential to maintaining grid stability, optimizing energy storage systems, and ensuring effective demand-response strategies.

To cope with this circumstance, the rapid advancement of Internet of Things (IoT) technology has transformed the landscape of energy management by enabling real-time data collection and processing. IoT-enabled smart grids can harness vast amounts of data in a distributed manner to make intelli-

gent localized energy management decisions [4]. These systems continuously monitor and analyze energy production, consumption, and distribution, allowing for proactive adjustments to enhance efficiency. However, traditional cloud-based processing approaches typically introduce latency and require significant computational resources. As the volume of energy data grows exponentially, the reliance on centralized cloud computing infrastructures can lead to bottlenecks, increased communication overhead, and security vulnerabilities resulting in management inefficiency. Edge computing has emerged as a promising solution to address these challenges by processing data closer to the source. By leveraging edge intelligence, smart grids can facilitate low-latency, real-time analytics, and localized decision-making, which are particularly beneficial for solar energy management. Deploying machine learning models directly on edge devices reduces dependency on remote servers, enabling faster response times and improved reliability.

Despite significant progress in machine learning and deep

learning methodologies for solar energy forecasting, existing models struggle with a trade-off between accuracy and computational efficiency [5]. Many conventional forecasting models fail to capture long-range temporal dependencies in time-series data, resulting in suboptimal long-term predictions [6], [7]. This limitation is particularly problematic for applications requiring extended forecasting horizons, such as solar energy planning and storage optimization. Additionally, deploying complex deep learning models in resource-constrained edge environments is challenging due to high computational and memory demands. To make this possible, we need models that are perfectly balanced between predicting what will happen and making good use of resources so that they can be used in real-life smart grid applications. Developing lightweight yet efficient deep learning frameworks tailored for edge computing environments is crucial to achieving scalable and effective solar energy forecasting solutions.

Motivated by the above analysis, this paper introduces an edge intelligence-driven hybrid deep learning model that integrates Temporal Convolutional Networks (TCN) and Gated Recurrent Units (GRU) to enhance solar energy forecasting accuracy. The main contributions of this paper include

- An edge intelligence-leveraged solar energy forecasting model (TGNet) has been proposed, which combines the advantages of TCN and GRU for accuracy enhancement. In addition, optimization strategies via neural architecture search, quantization, and neuron pruning have been integrated into the model to obtain low computational complexity while ensuring high accuracy for edge deployment on resource-constrained IoT devices.
- The proposed TGNet model has been validated using two well-known datasets, including Alibaba Competition and GEFCom, compared with existing algorithms. In terms of mean square error (MSE), mean absolute error (MAE), and low architectural complexity, numerical results show that the proposed model does better than others.

The remainder of this paper is organized as follows. Section II investigates existing work in solar energy forecasting and edge computing applications. Section III presents the proposed TGNet model and optimization techniques. Section IV details the experimental setup, datasets, and performance evaluation and analysis. Finally, Section V concludes the paper with future research directions.

II. RELATED WORK

In the domain of solar energy forecasting, hybrid deep learning models have shown significant promise in enhancing predictive accuracy and efficiency. For instance, the authors in [8] proposed an integrated model including the nonlinear autoregressive with exogenous inputs (NARX) component and long short-term memory (LSTM) networks to predict daily solar irradiance. This model effectively handles nonlinear relationships and long-term dependencies, achieving a mean absolute error and a root mean square error (RMSE)

that outperforms other benchmark models in accuracy. A predictive model for photovoltaic (PV) power utilizing the TCN and White Shark Optimization (WSO) algorithm enhances performance through feature selection based on the maximal information coefficient (MIC), thereby increasing the accuracy and reliability of PV power output predictions across various PV systems [9]. In [10], a model combining Convolutional Neural Networks (CNN) and LSTM has been proposed. The model integrated the variational mode decomposition (VMD) model and applied it for short-term PV power forecasting. This hybrid model provides better efficiency than other deep learning models at capturing both temporal and spatial dependencies, which makes solar power predictions more accurate. The Bidirectional Long Short-Term Memory (BiLSTM)-GRU embedded model incorporates dropout techniques proposed by authors in [11], outperforming other classical models in predicting solar irradiance. This model effectively captures multivariate data dependencies, achieving low RMSE and MAE values and indicating high prediction accuracy. In [12], a 1D CNN-GRU model has been built for short-term solar PV power forecasting. This model has the lowest RMSE and higher efficiency in the training phase than other models. The hybrid TCN-GRU model has been used for various purposes. In [13], the authors created a powerful hybrid TCN-GRU model for predicting traffic flow. As performance results, the model is better than baseline models such as LSTM, AutoRegressive Integrated Moving Average (ARIMA), and Support Vector Machine (SVM) in prediction accuracy. The study [14] crafted a cutting-edge multi-step wind speed prediction model employing a TCN-GRU hybrid. TCN was instrumental in capturing the complex features of wind speed time-series data, and GRU expertly managed multi-step forecasting. The study [15] utilized Grey Relational Analysis (GRA) to construct a spatial weight matrix by quantifying the correlation between air quality monitoring stations. This enabled the proposed GRA-GRU model to incorporate spatial dependencies into the prediction process, enhancing its performance. Nevertheless, the aforementioned studies exposed difficulty in deploying edge devices due to their complexity.

Deploying deep learning for solar energy forecasting on edge devices is a novel research area with limited studies. The Informer model provides long sequence forecasting with a ProbSparse self-attention mechanism, optimizing computation and memory. Its generative decoder enhances inference speed, crucial for latency-sensitive edge computing. However, the Informer struggles with non-stationary data, impacting accuracy in real-world scenarios [16]. The TimesNet effectively captures temporal patterns, which is beneficial for real-time analysis in edge computing. However, It may lack specific optimizations for constrained edge environments [17]. By combining TCNs and seasonal-trend decomposition, Autformer improves long-term sequence prediction and performance. However, its complexity can lead to higher computational overhead, complicating deployment on limited-resource edge devices [18]. With series rationalization and

de-stationary attention, the nonstationary transformer model handles nonstationarity well, making things more predictable. It significantly lowers MSE across transformer models, making it reliable for dynamic environments. However, the complexity of managing non-stationarity may increase computational demands, posing challenges in edge computing [19]. The DLinear model is lightweight and efficient, ideal for resource-constrained edge computing. However, they might need to capture complex temporal dependencies, potentially hindering forecasting accuracy effectively [20]. The GRU-XGBoost model [21], a hybrid approach, combines GRU for temporal dependencies with XGBoost for non-linear relationships, providing balanced time series forecasting. Its nature is advantageous for edge computing, delivering strong predictions with lower computational costs. However, the complexity of the model's integration might impede its deployment and maintenance on edge devices, which prioritize simplicity.

According to the above investigation, there has currently been limited comprehensive research on solar energy forecasting that considers both short-term and long-term perspectives, especially for edge implementation. The study [22] investigated the integration of CNN, LSTM, and Transformer (CNN-LSTM-TF model), demonstrating the potential of complex architectures in reducing forecasting errors. However, despite achieving high accuracy, the complexity of such tri-hybrid models often results in significant computational costs, highlighting the trade-off between accuracy and computational efficiency. Next, the work [23] proposed a two-stage solar power generation (SPG) model consisting of a CNN-BiLSTM encoder to handle multivariate time series (24 hours) and a Multilayer Perceptron (MLP) regression module. Although this architecture offers optimization advantages, the BiLSTM component significantly increases computational costs. Additionally, the research [24] proposed Hierarchical Temporal Convolutional Neural Network Architectures (HTCNN) A1 and A2 models utilizing TCN to forecast solar power output from distributed PV systems. This model has a complex architecture with multiple convolution stages and residual blocks, where the convolution stages use TCN to extract features from both aggregated data and data from individual sites. The residual blocks help the model learn deeper features, but they also increase the number of parameters to be trained. Although the model achieves high accuracy, it requires large computational and memory resources due to the complexity of its architecture with multiple layers and residual blocks, making the training and deployment of the model resource-intensive.

III. PROPOSED TGNET MODEL

A. SYSTEM STATEMENTS

In order to create an effective solar energy forecasting system, we approach the task as a machine learning challenge, where the input consists of multivariate time series data and the output is the predicted power generation time series. The input data undergoes preprocessing before being split into

training, validation, and test sets. Optimizing the model's hyperparameters is a crucial step to ensure accurate and efficient predictions. A TCN for feature extraction, a GRU for sequential modeling, and Neural Architecture Search (NAS) for automated model optimization make up the main parts of the system.

- **Input Data:** Historical solar energy generation data and meteorological characteristics, including sun irradiance, temperature, and wind velocity, are input data for the forecasting model.
- **Data Preprocessing:** The initial dataset is cleaned, normalized, and transformed to handle missing data, noise, and feature scaling to ensure its suitability for the neural network's learning mechanisms.
- **Data splitting:** It involves dividing the dataset into two sets: a testing set for an unbiased assessment of unknown data and a training set for parameter estimation.
- **Neural Architecture Search:** The NAS framework autonomously finds and builds the best neural network architecture, improving model design for predictive performance and edge deployment computing efficiency.
- **Model Training:** The TCN module extracts important features from multivariate time-series data. In this context, "multivariate" refers to the dependencies among the input features (solar radiation, temperature, wind speed, etc.) and between these inputs and the output (solar energy generation). At the same time, the Dense Layer prepares them for a GRU module, which emphasizes temporal dependencies and produces a dense layer that predicts solar energy generation.

Model evaluation is executed over cross-validation on the training set, facilitating the best model selection. The optimal model is then tested on the test set, comparing predicted results to actual values for performance assessment. Cross-validation iterations enhance the model and fine-tune parameters for superior predictive accuracy.

In particular, the input of the problem is defined as follows:

$$\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T\}, \quad (1)$$

where $\mathbf{X}$ represents the multivariate input time series, consisting of data points from $\mathbf{x}_1$ to $\mathbf{x}_T$. Each $\mathbf{x}_t \in \mathbb{R}^d$ is a feature vector at time step t , containing d variables related to solar energy efficiencies, such as solar radiation, temperature, and wind speed. The variable T denotes the length of the input sequence, representing historical data.

The output is defined as:

$$\mathbf{Y} = \{y_{T+1}, y_{T+2}, \dots, y_{T+H}\}, \quad (2)$$

where $\mathbf{Y}$ denotes the time series of solar energy output that we aim to predict, from y_{T+1} to y_{T+H} . y_t is the energy output value at time step t ; H is the prediction horizon indicating the number of future time steps.

The task is to establish a precise mapping function:

$$\mathbf{f} : \mathbb{R}^{T \times d} \rightarrow \mathbb{R}^H \quad (3)$$

where f is a mapping function from the input space $\mathbb{R}^{T \times d}$ (with T time steps and d features) to the output space $\mathbb{R}^H$ (representing H future time steps to predict).

The goal of the model is to find an accurate mapping function such that

$$\mathbf{Y} \approx \mathbf{f}(\mathbf{X}). \quad (4)$$

When the predicted values $f(X)$ are as close as possible to the actual values Y , the model achieves better predictive accuracy. This formulation leverages the temporal dependencies captured by the input sequence T , the multivariate relationships modeled by the feature dimension d , and the forecasting ability defined by the output horizon H .

B. THE PROPOSED MODEL ARCHITECTURE

The architecture of a TCN confidently comprises several 1D convolution layers with progressively more significant expansion coefficients. A nonlinear activation function (ReLU) that can also include a normalization layer (such as batch normalization) follows each concatenation layer. A TCN is constructed with residual blocks, each featuring two layers of dilated volumes that utilize shared weighting parameters. This design empowers TCN to possess an extensive receptive field and the capability to master intricate features while efficiently preserving a relatively compact number of parameters.

The proposed architecture is designed to handle multivariate relationships in solar energy forecasting data. In this context, "multivariate" encompasses both the dependencies among input features (solar radiation, temperature, wind speed, etc.) and the relationships between these inputs and the output (solar energy generation). Instead of treating each variable separately like in univariate models, our TCN-GRU architecture uses its convolutional operations to find cross-feature correlations. The TCN component's dilated convolutions effectively pull out temporal patterns across multiple variables simultaneously, while keeping their connections between them. This approach aligns with recent research [25] on multivariate deep learning for microgrid energy management but enhances it through the hierarchical feature extraction capabilities of TCN combined with GRU's sequential modeling strengths. The multivariate processing is particularly important in solar energy forecasting, where complex relationships exist between meteorological variables and their collective impact on energy generation.

To provide a clearer understanding of the multivariate time-series input and its relationship with vector-embedded connections, Figure 1 illustrates how the input features are structured. The figure shows how different variables (like solar radiation, temperature, and wind speed) change over time and how embeddings represent them. These embeddings are what the TCN uses to do feature extraction later on. The figure shows how each variable at each time step is defined as a vector and how these vectors are connected to form a multivariate time series.

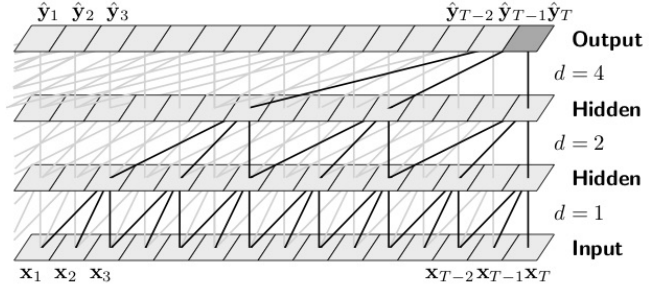


FIGURE 1: The relationships in input multivariate data and vector-embedded connections.

To efficiently capture temporal dependencies in time-series data, the TCN employs dilated convolutions. Standard convolutional layers work on time steps that are next to each other. Dilated convolutions, on the other hand, let the model process inputs with a spaced kernel, which exponentially increases the receptive field without adding more layers. This mechanism ensures that information from distant time steps can influence predictions, making it highly suitable for sequence modeling. We can mathematically express the dilated convolution operation in TCN as follows:

$$y[t] = \sum_{k=0}^{K-1} w_k \cdot x[t - d \cdot k] \quad (5)$$

where $y[t]$ is the output at time t , $x[t]$ is the input at time t , w_k represents the weight of the kernel at position k , K is the kernel size, and d is the dilation rate, which determines the spacing between selected input values.

To quantify the extent of information propagation, the receptive field is defined as:

$$\text{ReceptiveField} = 1 + \sum_{l=0}^{L-1} (K - 1) \cdot d_l \quad (6)$$

where L is the number of convolutional layers, and d_l represents the dilation rate at layer l . The dilation mechanism makes the receptive field bigger and bigger, which lets TCN capture long-term dependencies effectively while keeping the number of layers manageable.

Additionally, we incorporate residual connections to enhance gradient flow and mitigate the vanishing gradient problem. These connections enable stable training and improve model convergence:

$$o[t] = \text{Activation}(x[t] + F(x[t])) \quad (7)$$

where $o[t]$ is the output of the residual block at time t , $x[t]$ is the input to the residual block, and $F(x[t])$ represents the transformed features after convolution operations.

The proposed TCN-GRU architecture confidently incorporates the dilated convolution classes as follows:

$$\mathbf{h}_t^{(l)} = \sigma \left(\sum_{k=0}^{K-1} \mathbf{W}_k^{(l)} \cdot \mathbf{x}_{t-d \cdot k}^{(l-1)} + \mathbf{b}^{(l)} \right) \quad (8)$$

where $\mathbf{h}_t^{(l)}$ is the output of layer l at time t , $\mathbf{x}_t^{(l-1)}$ is the input of layer l (which is also the output of the previous layer), K is the kernel size, d is the dilation rate, and $\mathbf{W}_k^{(l)}$ and $\mathbf{b}^{(l)}$ are the weight and bias parameters of layer l , respectively. The function σ represents the activation function, which can be ReLU, sigmoid, or tanh.

The extracted temporal features from TCN are then passed to the GRU module for sequential modeling. Unlike traditional recurrent networks, the GRU employs a gating mechanism to control information flow. This allows it to selectively retain or discard past information, ensuring that relevant dependencies are preserved while reducing redundancy. The GRU update process follows this definition:

$$\mathbf{r}_t = \sigma(\mathbf{W}_r \cdot [\mathbf{h}_{t-1}, \mathbf{z}_t] + \mathbf{b}_r) \quad (9)$$

$$\mathbf{u}_t = \sigma(\mathbf{W}_u \cdot [\mathbf{h}_{t-1}, \mathbf{z}_t] + \mathbf{b}_u) \quad (10)$$

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_h \cdot [\mathbf{r}_t \odot \mathbf{h}_{t-1}, \mathbf{z}_t] + \mathbf{b}_h) \quad (11)$$

$$\mathbf{h}_t = (1 - \mathbf{u}_t) \odot \mathbf{h}_{t-1} + \mathbf{u}_t \odot \tilde{\mathbf{h}}_t \quad (12)$$

where $\mathbf{r}_t$ and $\mathbf{u}_t$ represent the reset and update gates, respectively, $\tilde{\mathbf{h}}_t$ is the candidate hidden state, and $\mathbf{h}_t$ is the final hidden state at time t . The reset gate determines how much past information should be discarded, while the update gate decides how much of the new candidate state should be retained.

The proposed TCN-GRU model can find both short-term and long-term dependencies in multivariate time series data by combining the sequential modeling strengths of GRU with the convolutional feature extraction strengths of TCN. This makes the model computationally efficient and suitable for real-time edge computing applications.

C. OPTIMIZATION PROCESS

To optimize the proposed model for deployment on resource-constrained edge devices, we leverage NAS, which is based on a reinforcement learning algorithm. The optimization process encompasses several critical computational steps, which are outlined below.

Firstly, to enhance the efficiency of the TCN, we incorporate the MobileNet (MBConv) block. This block is designed to minimize the number of parameters and computational costs, making it suitable for resource-constrained devices. By fine-tuning its hyperparameters, including the number of filters, kernel size, and dilation ratio, we identify the optimal configuration to balance model performance and efficiency.

After determining the hyperparameters of MBConv, the next phase concentrates on designing and searching for the

optimal architecture. This step involves using NAS, systematically exploring various configurations to identify the most effective model structure. The key parameters considered during this process include:

- Number of filters: 16 to 128 (step size: 16),
- Kernel size: 3 to 7 (step size: 2),
- Number of units in the GRU layer: 10 to 50 (step size: 10),
- Learning rate: 1e-2, 1e-3, 1e-4.

We employ the Hyperband algorithm at a factor of 3 and a maximum of 10 epochs per model to efficiently explore the search space. By dynamically allocating computational resources to promising configurations, the approach balances search efficiency and accuracy. We select architectures with the best validation loss, ensuring that the resulting model meets accuracy and efficiency requirements.

This reliance on validation loss is crucial, as it provides an unbiased metric for evaluating model performance on unseen data, helping to prevent overfitting. Furthermore, relying on validation metrics ensures that the selected architecture generalizes well to real-world scenarios, achieving a balance between predictive accuracy and computational efficiency.

Pruning and quantization are crucial for edge deployment as they reduce the model's size and computational demands. Pruning removes unimportant connections in the neural network, reducing the number of parameters and operations. Quantization lowers the accuracy of the model's weights and activations (for example, from 32-bit floating point to 8-bit integer). This makes the model's memory footprint smaller and speeds up inference on hardware that can do integer math. These techniques enable deploying complex models on resource-constrained edge devices, making real-time solar energy forecasting feasible.

D. FEATURE IMPORTANCE ANALYSIS

We did a SHAP (SHapley Additive Explanations) feature importance analysis for both short-term (15-step) and long-term (150-step) prediction horizons to learn more about how well the TGNNet model could make predictions. The analysis reveals which input features contribute most significantly to the model's predictions and how their relative importance shifts across different time horizons. Our analysis identified three dominant features that consistently influence the model's predictions:

- **VAR175 (0.00062)**: Remains the most important feature even for long-term predictions, likely representing a critical power generation or grid stability parameter.
- **VAR178 (0.00043)**: Emerges as the second most important feature, showing increased relative importance in longer-term forecasts compared to short-term results.
- **POWER (0.00036)**: Historical power values are important for long-term forecasting, demonstrating strong autoregressive characteristics.

The importance of features shifts noticeably when comparing short-term (15-step) and long-term (150-step) predictions,

as shown in Table 1:

The feature importance results reveal several noteworthy patterns:

- **Shifting Importance Balance:** The dominance of VAR175 decreases for longer-term predictions, from being $2.7\times$ more important than VAR178 in short-term forecasts to just $1.4\times$ in long-term forecasts.
- **Temporal Dynamics:** VAR175 appears more critical for short-term predictions, while VAR178 and POWER gain relative importance for longer-term forecasting.
- **Domain-Specific Insights:** The increasing importance of historical POWER values for longer horizons aligns with strong autocorrelation effects typically observed in power prediction systems.
- **Feature Synergy:** For long-term forecasts, the model relies more evenly on multiple features rather than being dominated by a single feature, suggesting that different variables capture complementary temporal dynamics.

This shifting feature importance between short-term (15-hour) and long-term (150-hour) horizons reveals fundamental differences in the temporal dynamics governing solar energy production. For short-term forecasting, immediate environmental conditions represented by VAR175 (surface thermal radiation down) dominate predictions because they directly influence imminent energy generation. However, as the prediction horizon extends to nearly a week, the model increasingly relies on broader patterns ... captured by VAR178 (top net solar radiation) and historical POWER values (see Table 1).

The increasing importance of VAR178 for long-term forecasts (+26%) suggests that upper-atmosphere radiation patterns, which reflect larger weather systems and atmospheric conditions, become more predictive over extended timeframes. This aligns with meteorological principles where macro-scale atmospheric phenomena become increasingly relevant for predictions beyond 72 hours.

Furthermore, our analysis uncovered important diurnal and seasonal effects in the feature importance patterns. Since each timestamp represents one hour, the 15-step forecast operates within typical daily cycles, while the 150-step forecast must account for multiple day-night transitions and potentially weekend-weekday variations. The model adapts to these patterns as follows:

- The importance of immediate radiation measurements (VAR175) decreases for long-term forecasts as daily cyclical patterns become more predictable from historical trends
- Historical POWER values gain importance (+29%) for long-term forecasts as they encapsulate both recent performance and seasonal patterns
- The relative weights of VAR178 increase proportionally to the forecast horizon, reflecting its value in capturing emerging weather patterns that manifest over several days

These results show that our TGNNet model changes how it uses features based on time scale, an important trait for making accurate predictions about the sun over a wide range of prediction horizons. The model can accurately predict both short-term environmental conditions and long-term weather patterns. This makes it a great choice for edge deployment in smart grid applications, where both forecasts are useful for making different operational decisions.

These findings validate that our TGNNet model successfully identifies and leverages the key drivers in energy forecasting that align with established research in the field. They also demonstrate how the model dynamically adjusts its reliance on different features based on the prediction horizon, contributing to its robust performance across both short-term and long-term forecasting scenarios.

IV. EXPERIMENTAL RESULTS AND DISCUSSIONS

A. DATASETS AND PREPROCESSING

a: Competition Dataset

Alibaba hosted the UNILAB Track 3 Algorithm Competition, Renewable Energy Production Prediction, where the first set of data originated. It includes environmental data about solar energy production at the same location for 300 days, from 6.am to 8.p.m. The dataset used in the study consists of 4500 records, representing information on factors affecting solar energy production. Each record contains data for a specific hour of the day, represented by two columns, "Day" and "Hour", with the data type being integer (int64). In addition to the time information, the dataset also records the wind direction ("Dir") and wind speed ("Spd"), which are two important factors affecting the performance of the solar energy system, along with the temperature ("Temp"). The "Radiation" column records the solar radiation level, which is a factor that directly determines solar energy production. This dataset was previously employed in [21] to develop predictive models for solar energy generation, demonstrating its effectiveness in time-series forecasting.

In the data preprocessing stage with this dataset, the features considered for inclusion in the forecasting model include wind direction ("Dir"), wind speed ("Spd"), temperature ("Temp") and radiation to ensure objective and accurate forecasting. The data is divided into 3 datasets, including a training set, validation set, and test set with a ratio of 70:15:15, ensuring continuity in terms of time.

In our data preprocessing stage, first, the raw data from three solar farms ("farm1", "farm2", and "farm3") are loaded, and unnecessary columns are removed. We then convert the "TIMESTAMP" column to date and time format and set it as the index for each dataset. Consequently, we use the StandardScaler method to normalize data. We calculate the weight for each data sample using the class weight function to enhance the impact of low-energy output samples during training. We convert data to a time series format with a fixed window length. We divide the examined data into a training set, a validation set, and a test set, adhering to the 70:15:15 ratio.

TABLE 1: Feature importance changes across different step evaluations.

Feature	15-Step Importance	150-Step Importance	Change
VAR175	0.00090	0.00062	-31%
VAR178	0.00034	0.00043	+26%
POWER	0.00028	0.00036	+29%

TABLE 2: Description of variables in the GEFCom2014 dataset

Variable name	Meaning	Units	Notes
ZONEID	Zone identifier	–	Identifies the geographic location or zone where data was recorded
TIMESTAMP	Timestamp	–	From which time features like dow, doy, hour, etc. can be extracted
VAR78	Total column liquid water (tclw)	kg m ⁻²	Total liquid water in atmospheric column (vertically integrated cloud liquid water)
VAR79	Total column ice water (tciv)	kg m ⁻²	Total ice water in atmospheric column (vertically integrated cloud ice water)
VAR134	Surface pressure (SP)	Pa	Atmospheric pressure at the ground surface
VAR157	Relative humidity at 1000 mbar (r)	%	Relative humidity at 1000 mbar level - determined by saturation value
VAR164	Total cloud cover (TCC)	(0–1)	Total cloud coverage ratio, derived from model cloud levels (overlapping assumption)
VAR165	10 metre U wind component (10u)	m s ⁻¹	Eastward wind component measured at 10m height
VAR166	10 metre V wind component (10v)	m s ⁻¹	Northward wind component measured at 10m height
VAR167	2 metre temperature (2T)	K	Air temperature measured at 2m height
VAR169	Surface solar rad down (SSRD)	J m ⁻²	Surface downward solar radiation (accumulated)
VAR175	Surface thermal rad down (STRD)	J m ⁻²	Surface downward thermal radiation (accumulated)
VAR178	Top net solar rad (TSR)	J m ⁻²	Net solar radiation at top of atmosphere (accumulated)
VAR228	Total precipitation (TP)	m	Total precipitation (including large-scale and convective precipitation) as accumulated value
POWER	Solar power	W	Target variable to be forecasted in the solar energy track

b: GEFCom2014

The second dataset is taken from the Solar data of the Global Energy Forecasting Competition 2014 (GEFCom2014). GEFCom14 provides 12 weather variables and their corresponding hourly solar energy production. The descriptive statistics of the GEFCom2014 dataset, as detailed in Table 2, illustrate the diverse meteorological inputs used for solar forecasting, ranging from surface pressure and cloud cover to directional wind components and multiple radiation variables. These variables are well-aligned with physical drivers of solar power generation, such as atmospheric clarity, humidity, and irradiance. This dataset was utilized in [26], where the authors applied it to develop advanced forecasting models for solar energy prediction.

In our data preprocessing stage, first, the raw data from three solar farms ("farm1", "farm2", and "farm3") are loaded, and unnecessary columns are removed. We then convert the "TIMESTAMP" column to date and time format and set it as the index for each dataset. Consequently, we use the StandardScaler method to normalize data. We calculate the weight for each data sample using the class weight function to enhance the impact of low-energy output samples during training. We convert data to a time series format with a fixed window length. We divide the examined data into a training set, a validation set, and a test set, adhering to the 70:15:15 ratio.

B. EVALUATED METRICS

The following list contains the main performance evaluation parameters for the proposed model. The following formula presents MSE.

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2, \quad (13)$$

Where y_i is the actual value, $\hat{y}_i$ is the predicted value, and N is the number of samples. MSE measures the mean square of the deviation between the predicted value and the actual value. It is sensitive to outliers and gives greater weight to large errors. A low MSE value indicates that the model is capable of predicting accurately, but it should be noted that MSE has units of the square of the original unit. The following formula presents MAE (Mean Absolute Error).

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|. \quad (14)$$

MAE measures the average magnitude of the prediction error and is less sensitive to outliers than RMSE. A low MAE indicates a model with good overall accuracy. R^2 score (coefficient of determination) is presented as.

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}, \quad (15)$$

Where $\bar{y}$ is the mean value of y_i . The R^2 score evaluates how well the model fits the data and shows how much of the variation in the dependent variable the model can account for. The closer the R^2 score is to 1, the better the model fits the data. Formulas (13) to (15) introduce the evaluation metrics used to assess the model's predictive performance. MSE and MAE provide quantitative measures of prediction error, with MSE emphasizing larger errors, while MAE offers a more interpretable average error magnitude. The R^2 score complements these by indicating the proportion of variance explained by the model, thereby offering a holistic view of the model's accuracy and reliability in forecasting solar energy output. These mathematical formulations make it possible for the model to accurately predict how much solar power will be produced by capturing temporal dependencies and improving performance through strong evaluation metrics.

TABLE 3: Optimized TCN-GRU Model Architecture

Layer	Parameters	Output Shape
Input Layer	(window_size, num_features)	(window_size, num_features)
TCN Block		
Conv1D (Block 1)	filters=num_features, kernel_size=1, padding="causal", activation='relu'	(window_size, num_features)
BatchNormalization (Block 1)	-	(window_size, num_features)
DepthwiseConv1D (Block 1)	kernel_size=3, dilation_rate=1, activation='relu', padding='same'	(window_size, num_features)
BatchNormalization (Block 1)	-	(window_size, num_features)
Dropout (Block 1)	rate=0.2	(window_size, num_features)
...		
Conv1D (Block 5)	filters=num_features, kernel_size=1, activation='relu', padding="causal"	(window_size, num_features)
BatchNormalization (Block 5)	-	(window_size, num_features)
DepthwiseConv1D (Block 5)	kernel_size=3, dilation_rate=16, activation='relu', padding='same'	(window_size, num_features)
BatchNormalization (Block 5)	-	(window_size, num_features)
Dropout (Block 5)	rate=0.2	(window_size, num_features)
GRU Block		
Dense	units=gru_unit	(window_size, gru_unit)
GRU	units=gru_unit	(window_size, gru_unit)
Dropout	rate=0.2	(window_size, predict_horizon)
Dense (Output Layer)	units=predict_horizon, activation='relu'	(window_size, predict_horizon)

C. EXPERIMENTAL SCENARIOS

To test how well the proposed model worked, four solar energy forecasting cases were made using the Competition and GEFCom2014 datasets: 15 steps (one day), 45 steps (three days), 90 steps (six days), and 150 steps (ten days). This setup adheres to the definition of long-term forecasting. The experiments were executed in a Python 3.10 environment on an RTX A4000 GPU. To ensure the reliability of the experimental results, all tests were performed under identical operational conditions. The experimental scenarios were designed to evaluate the proposed model comprehensively.

- **Performance Comparison.** This experiment is used to compare the forecasting performance of the model with other models across the three datasets. The evaluation metrics are RMSE, MAE, and the R^2 score against other state-of-the-art forecasting models.
- **Optimization Evaluation.** This scenario focuses on assessing the impact of the proposed optimization techniques on the proposed model, including optimized convolution blocks, neural architecture search, quantization, and neuron pruning techniques to reduce the model's size and complexity.
- **Sensitivity Analysis.** The goal of this experiment is to see how sensitive the proposed model is to changes in input factors and hyperparameters, such as the length of the input sequence and the model's hyperparameters. We fine-tune key hyperparameters of the model (e.g., number of layers, number of neurons, learning rate) to identify optimal configurations for forecasting performance. By analyzing the results (RMSE, MAE, R^2 score) obtained from adjusting these factors, we can determine the most critical factors influencing the model's performance and identify the optimal configuration for the proposed model.

D. RESULTS AND DISCUSSIONS

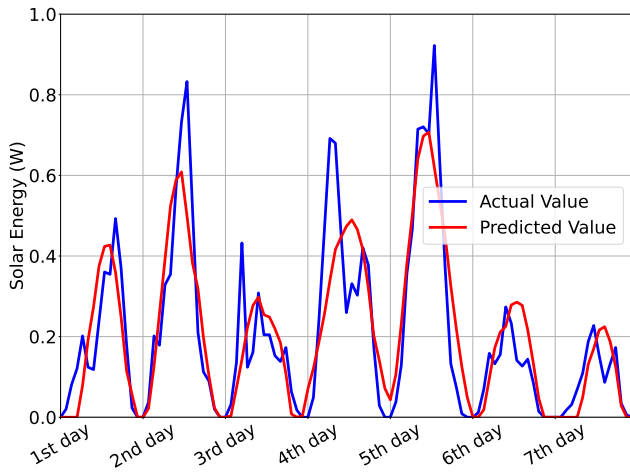
The parameter tuning for our proposed model is shown in Table 3.

To thoroughly evaluate the performance of the proposed model in light of the other state-of-the-art forecasting models, we undertook validation evaluations utilizing the same competing datasets together with the GEFCom2014 dataset. The parameters employed for comparison encompass MSE and MAE.

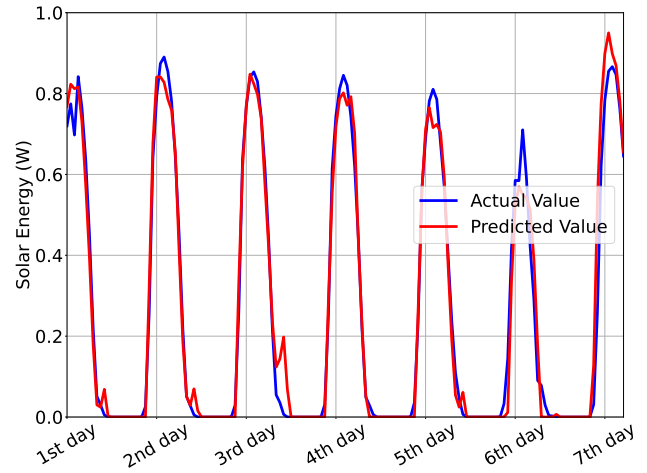
The results show that the proposed TNet model is more accurate than other models and can be used for both short-term (15 steps) and long-term (150 steps) forecasting purposes. Specifically, the MSE values obtained while forecasting with the competing dataset are 0.0063, 0.0058, 0.0056, and 0.0051, which correspond to the transitions from short-term to long-term scenarios. Likewise, in the context of the GEFCom2014 dataset, the MSE for the proposed model ranges from 0.0078 to 0.0044. Therefore, the proposed model's ability to adapt to two different datasets has supported the claim that it performs admirably in terms of its ability to generalize. This assertion is further corroborated by the MAE measurement parameter delineated in Table 4. **The trend of performance variation across different prediction horizons is visually depicted in Figure 3, illustrating the changes in MSE, MAE, and R^2 score.**

Based on the experimental results, the proposed model achieved its best predictive performance using the GEFCom2014 dataset with an input sequence of 150 time steps. The aim was to predict 150 corresponding output values, and the performance was evaluated using MSE, MAE, and R^2 , as presented in Table 5.

The results indicate that optimal performance was achieved with an input sequence length of 150, specifically with an MSE of 0.004404, MAE of 0.039380, and an accuracy R^2 of 94.11%. **While this represents the optimal performance**



(a) Forecasting results on the Alibaba Competition dataset.



(b) Forecasting results on the GEFCom2014 dataset.

FIGURE 2: Forecasting results of TGNNet on different datasets.

TABLE 4: Performance comparisons

Models	Metric ↓	Alibaba Competition				GEFCom2014			
		15	45	90	150	15	45	90	150
Informer ^[16]	MSE	0.012783	0.019615	0.103427	0.131893	0.027251	0.043939	0.032041	0.059199
	MAE	0.082409	0.104357	0.227575	0.251237	0.132939	0.138631	0.12632	0.178218
TimesNet ^[17]	MSE	0.014496	0.018866	0.027784	0.034643	0.010032	0.044495	0.045999	0.022512
	MAE	0.094709	0.109545	0.125303	0.132362	0.069683	0.136010	0.144405	0.102347
Autoformer ^[18]	MSE	0.030481	0.051724	0.075969	0.076494	0.023020	0.054124	0.08692	0.054124
	MAE	0.144049	0.162437	0.178979	0.201931	0.103717	0.154616	0.162872	0.217799
Non-stationary transformer ^[19]	MSE	0.024206	0.012479	0.027318	0.01803	0.030098	0.035854	0.033051	0.039763
	MAE	0.128303	0.081782	0.112632	0.203601	0.136945	0.121072	0.144224	0.144224
DLinear ^[20]	MSE	0.047359	0.045103	0.024422	0.037683	0.021998	0.030550	0.029522	0.027222
	MAE	0.159973	0.157069	0.117644	0.123949	0.126136	0.111983	0.109654	0.107475
GRU-XGBoost ^[21]	MSE	0.006905	0.024442	0.021118	0.027003	0.021817	0.028211	0.024474	0.020501
	MAE	0.066797	0.109514	0.104569	0.114723	0.098888	0.115150	0.109654	0.100644
TGNNet (proposed model)	MSE	0.0063	0.0058	0.0056	0.0051	0.007757	0.009267	0.005461	0.004404
	MAE	0.0556	0.0522	0.0517	0.0501	0.054802	0.055472	0.043656	0.039380

TABLE 5: The optimized input length of the forecasting model.

Model	Metric ↓	GEFCom2014				
		50	100	150	200	250
TGNNet	MSE	0.012208	0.009322	0.004404	0.005452	0.005568
	MAE	0.062308	0.055079	0.039380	0.043883	0.038456
	R^2	87.57%	90.04%	94.11%	92.57%	91.69%

found, Figure 3 provides a broader view of how these metrics generally evolve as the forecast horizon increases. However, when the input sequence length increased to 200 and 250, the R^2 value decreased, indicating no further improvement in predictive performance. This decline can be attributed to the increased amount of unnecessary information, which introduces noise and reduces the model's ability to generalize time series data effectively. Moreover, the lightweight model lacked the capacity to handle excessively long historical data.

TABLE 6: TCN-GRU Architecture

Layer		Output	Std	Dw
C1D-BN-A	C1D-BN-A-DW	(150, 64)	1,088	1,984
C1D-BN-A	C1D-BN-A-DW	(150, 96)	56,160	7,440
C1D-BN-A	C1D-BN-A-DW	(150, 128)	111,744	14,368
C1D-BN-A	C1D-BN-A-DW	(150, 128)	166,528	19,072
C1D-BN-A	C1D-BN-A-DW	(150, 96)	148,368	14,352
Dense	Dense	(150, 40)	3,880	3,880
GRU	GRU	(40)	9,840	9,840
Dense	Dense	(150)	6,150	6,150
Total			512,160	70,378

Note: C1D-BN-A denotes Conv1D with BatchNorm and Activation; C1D-BN-A-DW denotes the corresponding Depthwise Conv variant.

So, picking the right input sequence length is very important to make sure that the data is enough for the model to find important patterns without being hampered by noise, which will improve the accuracy of the predictions in the end.



FIGURE 3: TGNNet performance metrics across different prediction horizons.

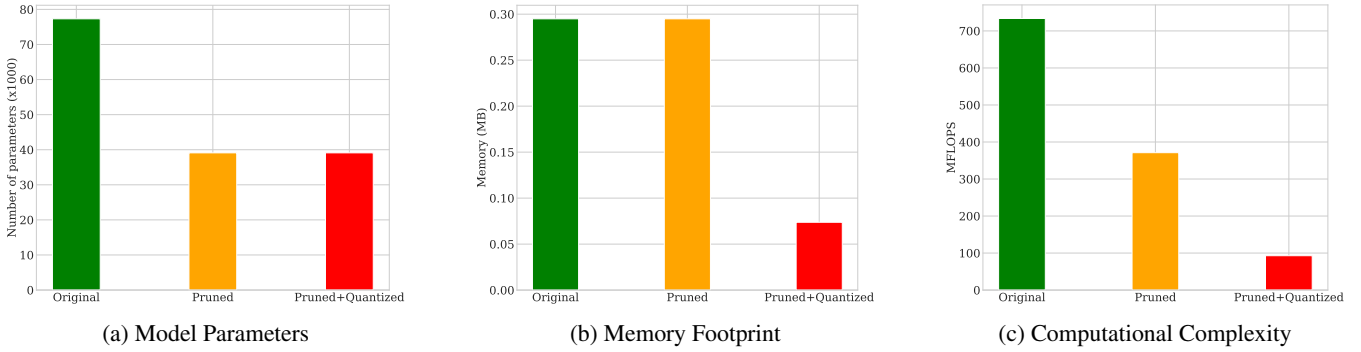


FIGURE 4: TGNNet model complexity.

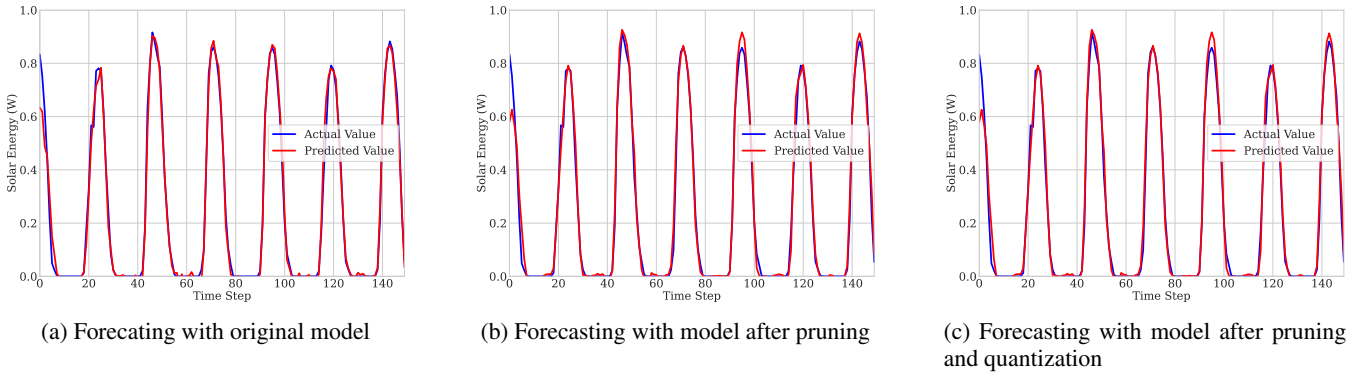


FIGURE 5: Pruned and quantized TGNNet performance.

After searching for the optimal architecture using NAS, pruning, and quantization, the TCN-GRU model achieves the structure shown in Table 6. Table 6 focuses on the architecture of the TCN-GRU model after optimization. This model uses both Conv1D and regular Conv1D depth layers. MBConv (MobileNet Convolution Block) is used to cut down on the number of parameters and the cost of computing them. Therefore, this makes the model more suitable for deployment on devices with limited resources, such as edge devices in IoT networks. Specifically, Table 6 compares the layers of the model along with the output shape and the corresponding number of parameters for each layer. The cumulative number of parameters for the model using conventional Conv1D is 512,160, while the model using deep Conv1D (Dw Conv)

only includes 70,378 parameters in Table 7. This finding indicates that MBConv's performance is more computationally efficient and requires less storage than traditional solutions, significantly reducing the model size without compromising prediction accuracy. The parameters presented in this table support the assertion that this optimization technique is important for efficient model execution on resource-constrained devices. A graphical comparison highlighting these reductions in parameters, memory footprint, and computational complexity (FLOPs) is provided in Figure 4.

Figures 2 presented the predictive outcomes of the TCN-GRU model as applied to test datasets derived from two distinct sources: the Competition Dataset (illustrated in Figure 2a) and the GEFCom2014 Dataset (depicted in Figure

TABLE 7: Optimization efficiency

Metrics	Conv	NAS+MBConv	Pruned	Pruned+Quant
MSE (↓)	0.0044	0.0009	0.0017	0.0017
MAE (↓)	0.0393	0.014	0.021	0.021
R^2 (↑)	94.11%	98.8%	97.7%	97.7%
Params (↓)	512,160	70,378	35,189	35,189
Mem footprint	-	0.29MB	0.15MB	0.07MB

2b). Both figures serve to compare the model's forecasted values against the actual (ground truth) values of solar energy production.

The TGNNet model's predictions on the competition dataset are shown in Figure 2a. They show a high level of accuracy for both short- and long-term hourly forecasts. Although the predicted values were very close to the actual measurements, this shows that the model is very good at capturing the time-dependent changes that happen with solar energy output. This observation suggests that the model excels not only in short-term predictions but also sustains a high degree of accuracy across extended intervals, thereby validating its robustness in the realm of long-term forecasting. In Figure 2b, concurrently, in the context of the GEFCom2014 dataset, the model exhibits commendable predictive accuracy, with the forecasted values closely mirroring the actual trends observed in solar energy output. The important aspect is that the data points in the figure show that the model correctly predicts the rising and falling trends in solar energy production over time. This proves that the model is stable and works well even when used with data from different solar farms. **Furthermore, the optimization techniques applied maintain this predictive capability, as illustrated by the sample forecasts shown in Figure 5 which compare the original and optimized model outputs.**

When pruning and quantization techniques are used, it becomes much easier to use the TCN-GRU model on edge devices with limited resources. Pruning, which strategically removes less important connections, achieved a sparsity of approximately 49.42%, effectively reducing the number of non-zero parameters from 70.37k to 35.19k. This reduction in parameters translates directly to a smaller memory footprint. Also, when the pruned model was quantized to 8-bit precision, it had a 4x compression ratio compared to the original FP32 model, which meant that only 0.07 MB of memory was needed. Importantly, the test loss remained relatively stable after pruning and quantization (0.0017), indicating that these optimizations achieved minimal impact on model accuracy. Figure 5 visually compares sample predictions from the original, pruned, and quantized models against the ground truth, further demonstrating the preserved forecasting quality. This balance between model size and accuracy is crucial for edge deployment, where computational resources and memory are limited. The reduced model size and computational complexity enable faster inference times and lower energy consumption, making the TCN-GRU model a practical solution for real-time solar energy forecasting on edge devices.

V. CONCLUSION AND DISCUSSION

This research introduces a hybrid edge intelligence framework for forecasting solar energy generation suitable for implementation on edge devices. The temporal extraction functions of TCN and the sequential data handling capabilities of GRU work better together in this hybrid framework, which makes both short-term and long-term forecasts more accurate. Empirical evaluations conducted on two well-known datasets indicate that the proposed framework exhibits superior performance compared to existing forecasting models. To focus on edge computing deployment, using the best methods like NAS, quantization, and model pruning reduces the architectural complexity while keeping the accuracy of the predictions high. This demonstration shows how flexible the framework is in edge computing settings. As we move forward with our plans, we will be adding more data sources, like satellite images and localized weather data, to improve the accuracy of our predictions and make the model more flexible in a range of settings.

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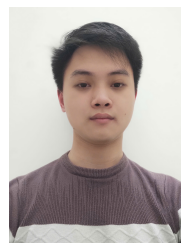
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